

A tensorial derivation of the DAD effect

Giacomo Po

July 30, 2026

Abstract

In this document we derive a correction to the sink strength of a dislocation loop under anisotropic diffusion. This correction is simply

$$\frac{\text{tr}(\mathbf{D}) + \mathbf{a}^T \mathbf{D} \mathbf{a}}{4\bar{D}}, \quad (1)$$

where $\mathbf{D}$ is the anisotropic diffusion tensor, and $\mathbf{a}$ is the axis of the loop. Due to its tensorial nature, this correction applies universally to any loop orientation and degree of anisotropy of the diffusion coefficient.

1 Woo's reaction coefficient α

Woo [1988] considers the reaction between two reacting species A and B of the form



where at least one of the species is diffusable and the reaction product is C. The rate of change caused by reaction (2) in the spatially averaged concentrations C_A , and C_B (in atomic fractions), is completely governed by the **reaction coefficient** α as follows

$$\dot{C}_A = \dot{C}_B = -\bar{D}\alpha C_A C_B, \quad (3)$$

where $\bar{D} = \det^{1/3}(\mathbf{D})$ is the average diffusion coefficient and $\mathbf{D} = \mathbf{D}_A + \mathbf{D}_B$ is the sum of the diffusion coefficients of species A and B.

If, in reaction (2), B is an indestructible sink, then the product αC_B , is equal to the usual sink strength k_B^2 , that is

$$k_B^2 = \alpha C_B. \quad (4)$$

The reaction coefficient is defined as the integral

$$\alpha = \frac{1}{\Omega} \int \nabla_{\xi} \mu \cdot d\mathbf{S}_{\xi}, \quad (5)$$

where Ω is the atomic volume, μ is the chemical potential of the pair probability density, and S_{ξ} is the oriented surface area of the **reaction volume**, defined in the coordinate system

$$\boldsymbol{\xi} = (\bar{D}\mathbf{D}^{-1})^{\frac{1}{2}} \mathbf{x}. \quad (6)$$

By introducing this coordinate transformation, Woo's original purpose was to reduce the anisotropic diffusion problem to a simpler isotropic diffusion problem in a modified coordinate system, with a transformed reaction volume that has a different (transformed) shape.

We now want to obtain the definition of α in real space. This will facilitate the tensorial calculation of the DAD correction in section 2. To do so, we see that, from (6), we have $d\boldsymbol{\xi} = \mathbf{F} d\mathbf{x}$, where $\mathbf{F} = (\overline{\mathbf{D}}\mathbf{D}^{-1})^{\frac{1}{2}}$ is the transformation Jacobian matrix. Hence the transformation rule for gradients and area vectors, are respectively,

$$\boldsymbol{\nabla}_{\boldsymbol{\xi}} = \mathbf{F}^{-T} \boldsymbol{\nabla} = \left(\frac{\mathbf{D}}{\overline{\mathbf{D}}} \right)^{\frac{1}{2}} \boldsymbol{\nabla}, \quad (7)$$

$$d\mathbf{S}_{\boldsymbol{\xi}} = \det(\mathbf{F}) \mathbf{F}^{-T} d\mathbf{S} = \left(\frac{\mathbf{D}}{\overline{\mathbf{D}}} \right)^{\frac{1}{2}} d\mathbf{S}. \quad (8)$$

Substituting these into (5), and using the symmetry of $\mathbf{D}$, we obtain

$$\alpha = \frac{1}{\Omega} \int \left(\frac{\mathbf{D}}{\overline{\mathbf{D}}} \right)^{\frac{1}{2}} \boldsymbol{\nabla} \mu \cdot \left(\frac{\mathbf{D}}{\overline{\mathbf{D}}} \right)^{\frac{1}{2}} d\mathbf{S} = \frac{1}{\Omega \overline{\mathbf{D}}} \int \mathbf{D} \boldsymbol{\nabla} \mu \cdot d\mathbf{S} \quad (9)$$

Since $-\mathbf{D} \boldsymbol{\nabla} \mu = \mathbf{J}$ is the flux, we introduce the **modified reaction coefficient**

$$\overline{\alpha} = \overline{\mathbf{D}} \alpha = \frac{1}{\Omega} \int \mathbf{D} \boldsymbol{\nabla} \mu \cdot d\mathbf{S}, \quad (10)$$

which clearly represents the total inward flow into the real-space reaction volume.

2 The modified reaction constant for a dislocation loop

Consider a dislocation loop with axis $\mathbf{a}$ orthogonal to the loop plane, and line tangent $\mathbf{t}$, as shown in Fig. 1. We assume that the real-space reaction volume for this loop is a slender tube with radius r . When the radius of the tube is small compared to the curvature of the loop, the modified reaction constant can then be computed as

$$\overline{\alpha} = \int \frac{d\overline{\alpha}}{d\ell} d\ell \quad (11)$$

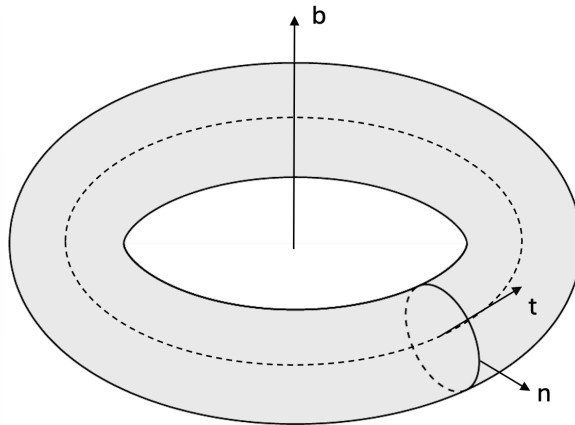


Figure 1: Dislocation loop and its toroidal reaction volume

where $\frac{d\bar{\alpha}}{d\ell}$ is the modified reaction constant per unit length, computed on a poloidal circle with unit outward normal $\mathbf{n}$. Hence,

$$\frac{d\bar{\alpha}}{d\ell} = \frac{r}{\Omega} \oint \mathbf{D} \nabla \mu \cdot \mathbf{n} d\theta. \quad (12)$$

The gradient of the chemical potential is due to the difference in chemical potential between the defect (dislocation line) and the bulk at infinity, $\Delta\mu$, and, by symmetry reasons, it is assumed to also be in the direction of $\mathbf{n}$, with a characteristic length L . Hence we obtain

$$\frac{d\bar{\alpha}}{d\ell} = \frac{r\Delta\mu}{L\Omega} \int_0^{2\pi} \mathbf{D} \mathbf{n} \cdot \mathbf{n} d\theta = \frac{r\Delta\mu}{L\Omega} \int_0^{2\pi} \mathbf{n}^T \mathbf{D} \mathbf{n} d\theta. \quad (13)$$

The outward unit normal $\mathbf{n}$ can be obtained as the rotation of the axis vector $\mathbf{a}$ about the tangent vector $\mathbf{t}$, hence

$$\mathbf{n} = \mathbf{R}(\mathbf{t}, \theta) \mathbf{a} \quad (14)$$

where $\mathbf{R}(\mathbf{t}, \theta)$ is a rotation matrix. This can be obtained from Rodrigues rotation formula

$$\mathbf{R}(\mathbf{t}, \theta) = \mathbf{I} + \sin \theta \mathbf{T} + (1 - \cos \theta) \mathbf{T}^2 \quad (15)$$

with $\mathbf{T}$ being the skew-symmetric axial matrix with the property $\mathbf{T} \mathbf{a} = \mathbf{t} \times \mathbf{a}$. With this notation, (16) becomes

$$\frac{d\bar{\alpha}}{d\ell} = \frac{r\Delta\mu}{L\Omega} \mathbf{a}^T \int_0^{2\pi} (\mathbf{I} + \sin \theta \mathbf{T} + (1 - \cos \theta) \mathbf{T}^2)^T \mathbf{D} (\mathbf{I} + \sin \theta \mathbf{T} + (1 - \cos \theta) \mathbf{T}^2) d\theta \mathbf{a}. \quad (16)$$

Note that, because of the integration in $[0, 2\pi]$, all terms linear in $\sin \theta$, $\cos \theta$, and $\sin \theta \cos \theta$, vanish. This leaves us with

$$\begin{aligned} \frac{d\bar{\alpha}}{d\ell} &= \frac{r\Delta\mu}{L\Omega} \mathbf{a}^T \int_0^{2\pi} (\mathbf{D} + \mathbf{T}^{2T} \mathbf{D} + \sin^2 \theta \mathbf{T}^T \mathbf{D} \mathbf{T} + \mathbf{D} \mathbf{T}^2 + (1 - \cos \theta)^2 \mathbf{T}^{2T} \mathbf{D} \mathbf{T}^2) d\theta \mathbf{a}. \\ &= \frac{r\Delta\mu}{L\Omega} \mathbf{a}^T (2\pi \mathbf{D} + 2\pi \mathbf{T}^{2T} \mathbf{D} + \pi \mathbf{T}^T \mathbf{D} \mathbf{T} + 2\pi \mathbf{D} \mathbf{T}^2 + 3\pi \mathbf{T}^{2T} \mathbf{D} \mathbf{T}^2) \mathbf{a}. \end{aligned} \quad (17)$$

Now, note that $\mathbf{T}^2 \mathbf{a} = \mathbf{t} \times \mathbf{t} \times \mathbf{a} = -\mathbf{a}$, so

$$\begin{aligned} \frac{d\bar{\alpha}}{d\ell} &= \frac{\pi r \Delta\mu}{L\Omega} ((\mathbf{T} \mathbf{a})^T \mathbf{D} (\mathbf{T} \mathbf{a}) + \mathbf{a}^T \mathbf{D} \mathbf{a}) \\ &= \frac{\pi r \Delta\mu}{L\Omega} ((\mathbf{t} \times \mathbf{a})^T \mathbf{D} (\mathbf{t} \times \mathbf{a}) + \mathbf{a}^T \mathbf{D} \mathbf{a}). \end{aligned} \quad (18)$$

2.1 Circular loop

For a circular loop, of radius R , $d\ell = R d\phi$, and the tangent vector can be obtained as the rotation of an arbitrary in-plane direction $\mathbf{p}$, about $\mathbf{a}$. Thence, using Rodrigues rotation formula again, we have

$$\begin{aligned} \mathbf{t} &= (\mathbf{I} + \sin \phi \mathbf{A} + (1 - \cos \phi) \mathbf{A}^2) \mathbf{p} \\ &= (\mathbf{I} + \sin \phi \mathbf{a} \times \mathbf{p} + (1 - \cos \phi) \mathbf{a} \times \mathbf{a} \times \mathbf{p}) \\ &= (\mathbf{p} + \sin \phi \mathbf{a} \times \mathbf{p} - (1 - \cos \phi) \mathbf{p}) \end{aligned} \quad (19)$$

The modified reaction constant therefore becomes

$$\begin{aligned}
\bar{\alpha} &= \frac{\pi r R \Delta \mu}{L \Omega} \left[\int_0^{2\pi} ((\mathbf{p} + \sin \phi \mathbf{a} \times \mathbf{p} - (1 - \cos \phi) \mathbf{p}) \times \mathbf{a})^T \mathbf{D} ((\mathbf{p} + \sin \phi \mathbf{a} \times \mathbf{p} - (1 - \cos \phi) \mathbf{p}) \times \mathbf{a}) d\phi \right. \\
&\quad \left. + 2\pi \mathbf{a}^T \mathbf{D} \mathbf{a} \right] \\
&= \frac{\pi r R \Delta \mu}{L \Omega} \left[\int_0^{2\pi} (\sin \phi \mathbf{p} + \cos \phi \mathbf{p} \times \mathbf{a})^T \mathbf{D} (\sin \phi \mathbf{p} + \cos \phi \mathbf{p} \times \mathbf{a}) d\phi + 2\pi \mathbf{a}^T \mathbf{D} \mathbf{a} \right] \\
&= \frac{\pi^2 r R \Delta \mu}{L \Omega} [\mathbf{p}^T \mathbf{D} \mathbf{p} + (\mathbf{p} \times \mathbf{a})^T \mathbf{D} (\mathbf{p} \times \mathbf{a}) + 2\mathbf{a}^T \mathbf{D} \mathbf{a}] \tag{20}
\end{aligned}$$

Finally, noting that $\mathbf{p}^T \mathbf{D} \mathbf{p} + (\mathbf{p} \times \mathbf{a})^T \mathbf{D} (\mathbf{p} \times \mathbf{a}) = \text{tr}(\mathbf{D}) - \mathbf{a}^T \mathbf{D} \mathbf{a}$, we obtain the final result independent of the arbitrary vector $\mathbf{p}$ as¹

$$\bar{\alpha} = \frac{\pi^2 r R \Delta \mu}{L \Omega} [\text{tr}(\mathbf{D}) + \mathbf{a}^T \mathbf{D} \mathbf{a}] . \tag{21}$$

In the isotropic case $\mathbf{D} = \bar{D} \mathbf{I}$ and the modified reaction constant becomes

$$\bar{\alpha}_0 = \frac{\pi^2 r R \Delta \mu}{L \Omega} 4\bar{D} = \frac{S}{\Omega} \frac{\Delta \mu}{L} \bar{D} \tag{22}$$

Therefore, the anisotropic version can be expressed as

$$\bar{\alpha} = \bar{\alpha}_0 \frac{\text{tr}(\mathbf{D}) + \mathbf{a}^T \mathbf{D} \mathbf{a}}{4\bar{D}} \tag{23}$$

The term $\frac{\text{tr}(\mathbf{D}) + \mathbf{a}^T \mathbf{D} \mathbf{a}}{4\bar{D}}$ provides a tensorial correction to the reaction constant, and therefore to the sink strength, accounting for the orientation of the loop ($\mathbf{a}$), and the arbitrary degree of the anisotropy of the diffusion tensor $\mathbf{D}$, which are responsible for the DAD effect.

References

CH Woo. Theory of irradiation deformation in non-cubic metals: effects of anisotropic diffusion. *Journal of Nuclear Materials*, 159:237–256, 1988. ¹

¹Note that, for a perfect torus, this equation is exact even when dropping the small tube approximation.